

Q1. Design Decision: Diverging colour scales [10 marks]

An analyst maps the change in rainfall from the 30-year average (mm) of many districts to colour on a map/heat map. Values range from -120 to +150.

- (a) Which type of colour scale is appropriate? What value should be the midpoint and why? [3]
- (b) Name a suitable R palette or give the two end colours and the neutral colour. [3]
- (c) Explain why a rainbow palette is a poor choice here. [2]
- (d) Sketch the legend (colour bar) with at least three labelled tick values. [2]

(a) Appropriate colour scale and midpoint [3]

Answer:

- Use a **diverging colour scale** because rainfall change can be both negative and positive.
- The **midpoint should be 0** because 0 represents **no change**.
- Negative values can represent decreased rainfall, while positive values represent increased rainfall.

Example:

Blue ← Negative change | 0 | Positive change → Red

(b) Suitable R palette / colours [3]

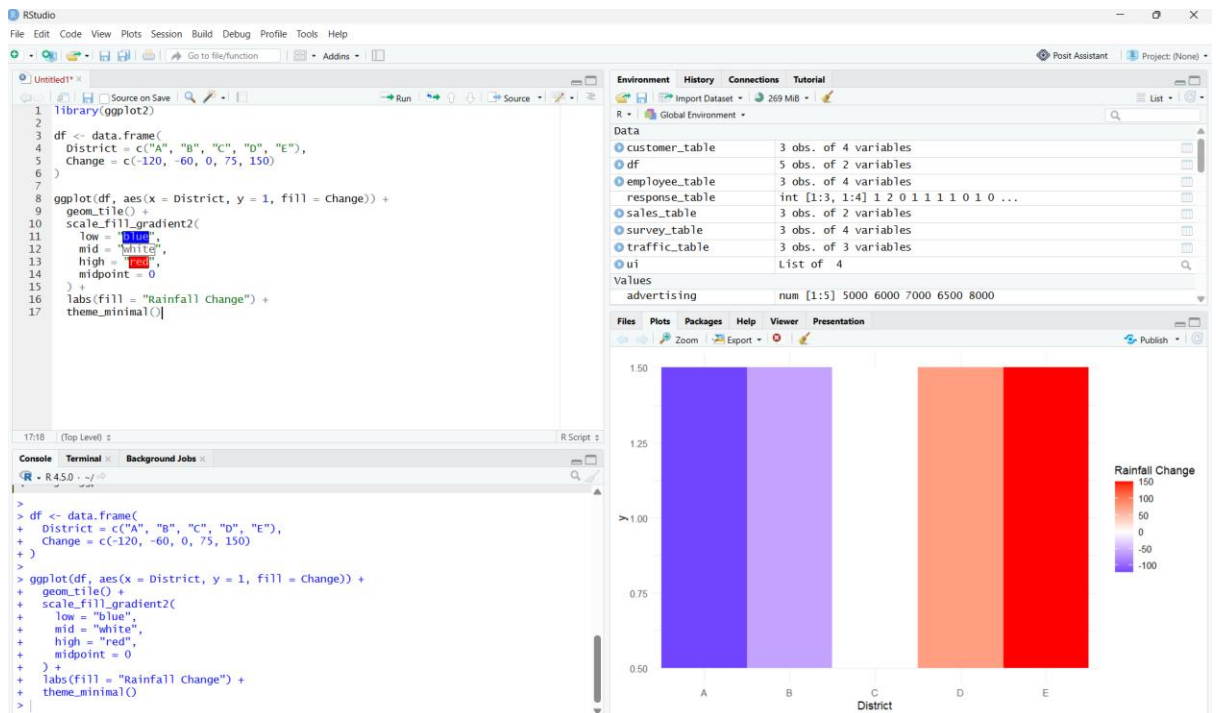
A suitable palette is **Red–Blue (RdBu)**.

- Negative values: **Blue**
- Midpoint (0): **White**
- Positive values: **Red**

R :

```
scale_fill_gradient2(  
  low = "blue",  
  mid = "white",  
  high = "red",  
  midpoint = 0  
)
```

Implementation:



(c) Why is a rainbow palette a poor choice? [2]

A rainbow palette is poor because:

1. The colours are **not perceptually uniform**, so equal numerical changes may not look equally important.
2. It does not clearly show the important **neutral midpoint (0)**.

(d) Sketch the legend [2]

The range is **-120 to +150**, so draw:

Rainfall Change

Blue White Red

| | |

-120 0 +150

You can label intermediate values such as:

-120 -60 0 +75 +150

|-----|-----|-----|-----|

Blue White Red

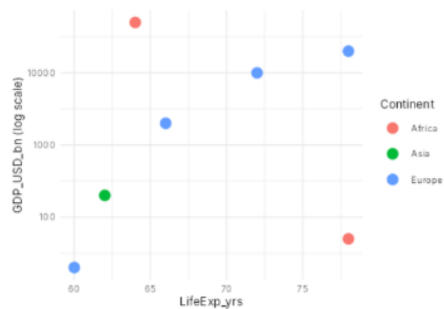
Q2. Visual to R Code: Aesthetics and a log axis [10 marks]

Name	LifeExp_yrs	GDP_USD_bn	Continent
India	62	200	Asia
Brazil	78	20000	Europe
Nepal	64	50000	Africa
Italy	60	20	Europe
Egypt	66	2000	Europe
Peru	72	10000	Europe
Chile	78	50	Africa

Write an R (ggplot2) program that reproduces the chart from the table.

(a) Write the complete R program (data frame creation and ggplot call). [7]

(b) State the aesthetic used for each variable, and why GDP_USD_bn needs a nonlinear axis. [3]



Country LifeExp_yrs GDP_USD_bn Continent

India	62	2000	Asia
Brazil	78	20000	Europe
Nepal	64	50000	Africa
Italy	60	20	Europe
Egypt	66	200	Europe
Peru	72	10000	Europe
Chile	78	50	Africa

(a) Complete R program [7]

```

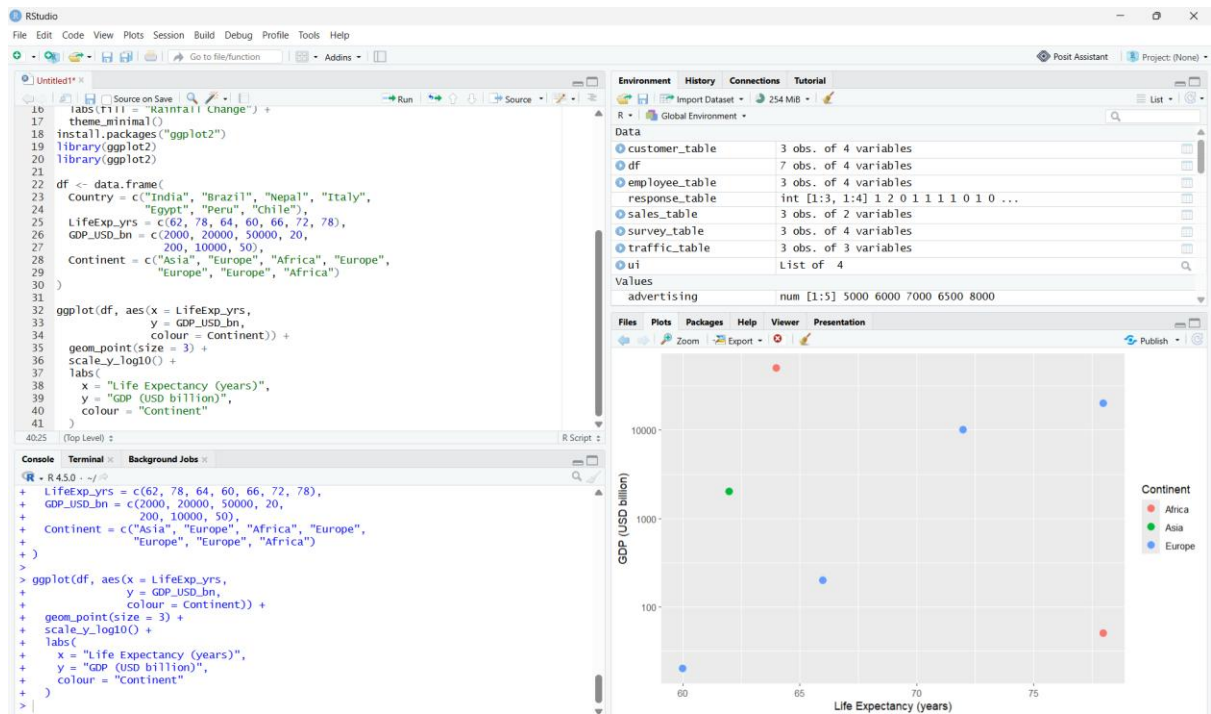
library(ggplot2)

df <- data.frame(
  Country = c("India", "Brazil", "Nepal", "Italy",
    "Egypt", "Peru", "Chile"),
  LifeExp_yrs = c(62, 78, 64, 60, 66, 72, 78),
  GDP_USD_bn = c(2000, 20000, 50000, 20, 200, 10000, 50),
  Continent = c("Asia", "Europe", "Africa", "Europe",
    "Europe", "Europe", "Africa")
)

ggplot(df, aes(x = LifeExp_yrs,
  y = GDP_USD_bn,
  colour = Continent)) +
  geom_point(size = 3) +
  scale_y_log10() +
  labs(
    x = "Life Expectancy (years)",
    y = "GDP (USD billion)",
    colour = "Continent"
  )

```

Implementation:



(b) Aesthetics and why GDP needs a nonlinear axis [3]

Aesthetics:

- $x = \text{LifeExp_yrs} \rightarrow \text{x-axis}$
- $y = \text{GDP_USD_bn} \rightarrow \text{y-axis}$
- $\text{colour} = \text{Continent} \rightarrow \text{colour}$

Why log scale for GDP?

GDP values range from **20 to 50,000**, which covers several orders of magnitude. A **logarithmic scale** makes both small and large GDP values visible.

`scale_y_log10()`

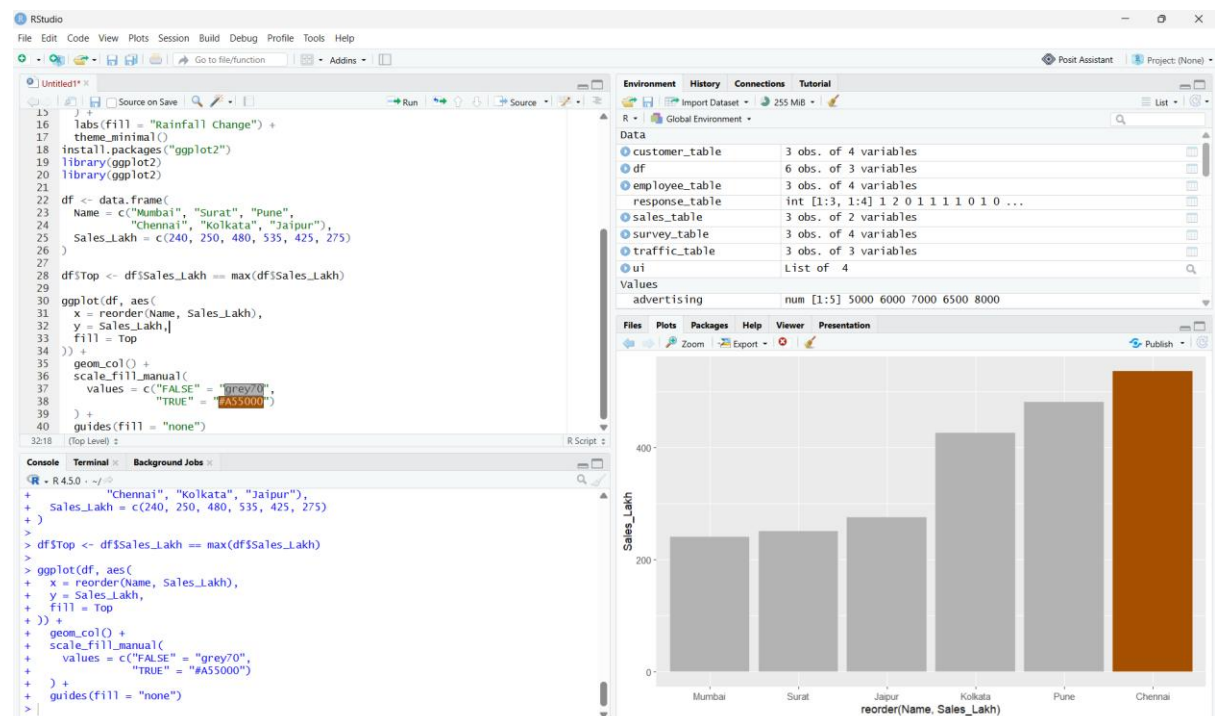
Q3. R Code to Visual: Colour as a tool to highlight [10 marks]

```
library(ggplot2)
df <- data.frame(
  Name = c("Mumbai", "Surat", "Pune",
    "Chennai",
    "Kolkata", "Jaipur"),
  Sales_Lakh = c(240, 250, 480, 535, 425, 275)
)
df$Top <- df$Sales_Lakh == max(df$Sales_Lakh)
ggplot(df, aes(x = reorder(Name, Sales_Lakh),
  y = Sales_Lakh, fill = Top)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c("FALSE" =
    "grey70", "TRUE" = "#D55E00")) +
  guides(fill = "none")
```

Read the R program carefully. Do not run it.

- Sketch the chart produced. Show bar order, colours and axes. [5]
- State the aesthetics, scale types and coordinate system used. [3]
- What is the role of the Top column? What would change if fill = Name were used? [2]

Implementation:



```
library(ggplot2)
```

```
df <- data.frame(
```

```
  Name = c("Mumbai", "Surat", "Pune",
```

```
    "Chennai", "Kolkata", "Jaipur"),
```

```
  Sales_Lakh = c(240, 250, 480, 535, 425, 275)
```

```
)
```

```
df$Top <- df$Sales_Lakh == max(df$Sales_Lakh)
```

```
ggplot(df, aes(x = reorder(Name, Sales_Lakh),
```

```

    y = Sales_Lakh,
    fill = Top)) +
  geom_col() +
  scale_fill_manual(values = c("FALSE" = "grey70",
    "TRUE" = "#A55000")) +
  guides(fill = "none")

```

(a) Sketch the chart

First arrange the cities in **ascending sales order** because of:

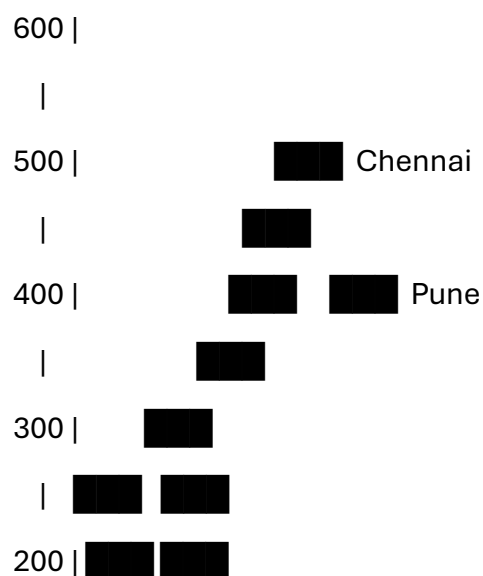
```
reorder(Name, Sales_Lakh)
```

Order:

1. Mumbai → 240
2. Surat → 250
3. Jaipur → 275
4. Kolkata → 425
5. Pune → 480
6. Chennai → 535

Your sketch should look approximately like:

Sales



+-----

Mumbai Surat Jaipur Kolkata Pune Chennai

All bars are **grey except Chennai**, which is highlighted in brown.

(b) Aesthetics, scales and coordinate system [3]

Aesthetics:

- `x = reorder(Name, Sales_Lakh)` → city names on x-axis
- `y = Sales_Lakh` → sales on y-axis
- `fill = Top` → colour indicates whether the city has maximum sales

Scale:

FALSE → grey

TRUE → brown

Coordinate system:

- Standard **Cartesian coordinates**
- x-axis = categorical city names
- y-axis = sales

(c) Role of Top and what happens with `fill = Name` [2]

Top is a **logical variable** that identifies the city having the maximum sales.

```
df$Top <- df$Sales_Lakh == max(df$Sales_Lakh)
```

Therefore:

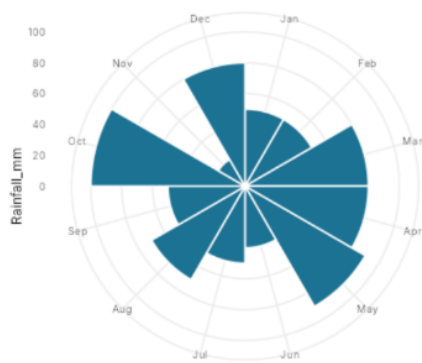
- TRUE → Chennai → highlighted
- FALSE → all other cities

If we use:

`fill = Name`

then **each city gets a different colour** instead of highlighting only the maximum.

Q4. Visual to Table: Reading a polar chart [10 marks]



The polar bar chart shows rainfall (mm) for 12 months. Radial grid lines are at 0, 20, 40, ... 100.

(a) Reconstruct the table (Month, Rainfall_mm). [5]

(b) Name the peak and the lowest month, and give the angle between adjacent bars. [3]

(c) State one advantage and one drawback of using polar coordinates here. [2]

a) Reconstruct the table [5]

Read the height of each bar using the radial grid.

Month	Rainfall_mm
January	50
February	40
March	60
April	60
May	70
June	40
July	30
August	50
September	30
October	70
November	80
December	90

So:

January = 50

February = 40

March = 60

April = 60

May = 70

June = 40

July = 30

August = 50

September = 30

October = 70

November = 80

December = 90

(b) Highest, lowest and angle between adjacent bars [3]

Highest rainfall:

- **December = 90 mm**

Lowest rainfall:

- **July and September = 30 mm**

There are 12 months, so each bar has:

$$\frac{360^\circ}{12} = 30^\circ$$

Therefore, the angle between adjacent bars is **30°**.

(c) One advantage and one drawback [2]

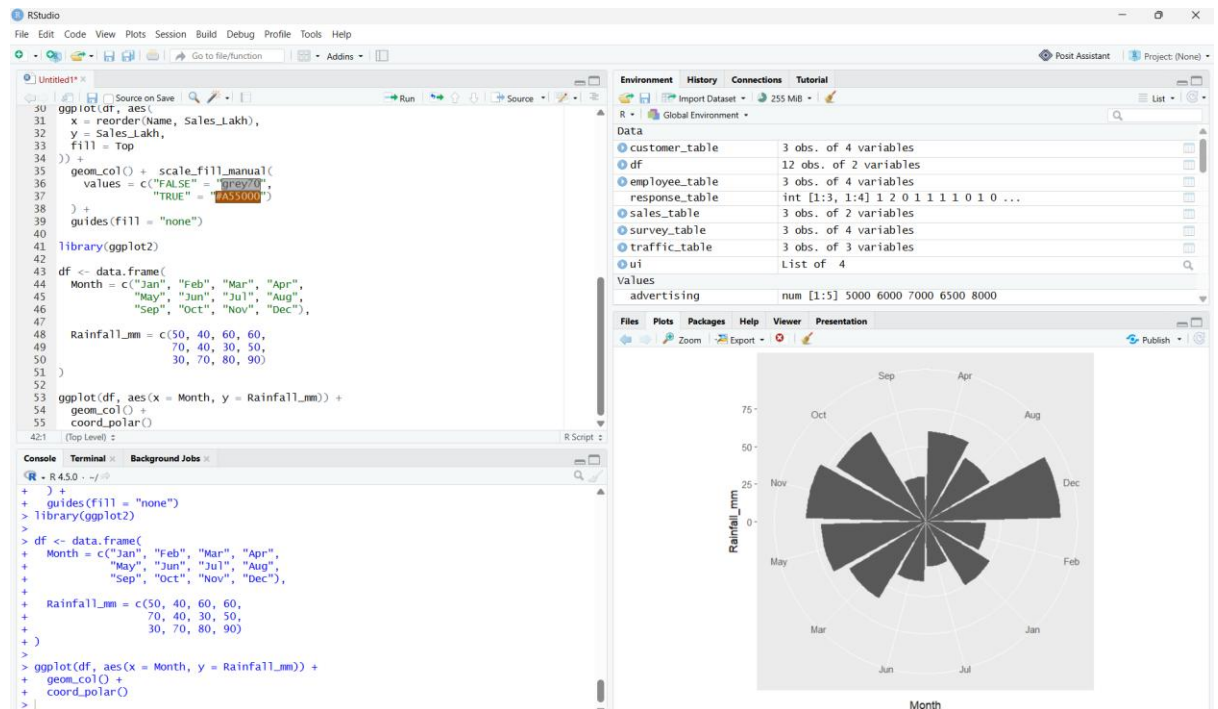
Advantage:

- Polar coordinates show the **cyclic nature of months/year** clearly.

Drawback:

- Comparing bar lengths is more difficult than in a normal Cartesian bar chart because the bars are arranged around a circle.

Implementation:



Q5. Table to Visual: Mapping data onto aesthetics [10 marks]

ID	StudyHours	Score	Dept	Attendance
S1	10	95	CSE	80
S2	5	80	ECE	90
S3	2	60	MECH	80
S4	2	85	MECH	70
S5	4	75	ECE	90
S6	2	80	CSE	80
S7	9	85	MECH	90
S8	8	80	MECH	70

The table records 8 students. Design one visualization that shows all four variables (ignore the ID column).

- State the data type (nominal / ordinal / continuous / discrete) of StudyHours, Score, Dept and Attendance. [2]
- Choose a suitable aesthetic for each of the four variables and name the scale type used. [3]
- Sketch the chart. Label the axes and draw the legends. [3]
- Explain why mapping Attendance to shape would be a poor choice. [2]

Given:

ID StudyHours Score Dept Attendance

S1	10	95	CSE	80
S2	5	80	ECE	90
S3	2	60	MECH	80
S4	2	85	MECH	70
S5	4	75	CSE	90
S6	2	80	CSE	80
S7	9	85	MECH	90
S8	8	80	MECH	70

(a) Data types [2]

Variable Data type

StudyHours Continuous

Score Continuous

Dept Nominal / categorical

Attendance Continuous

Note: Study hours are treated as continuous because hours are a measurable quantity.

(b) Suitable aesthetic for all four variables [3]

A good visualization is a **scatter plot**.

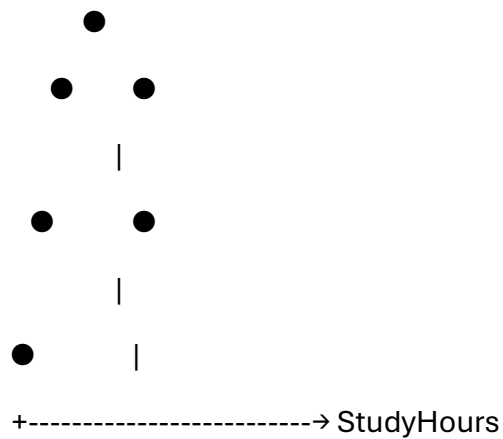
Use:

- **x-axis → StudyHours**
- **y-axis → Score**
- **colour → Dept**
- **size → Attendance**

Example:

Score

↑



Colour = Department

Size = Attendance

Aesthetic mapping

Variable	Aesthetic
StudyHours	x-position
Score	y-position
Dept	Colour
Attendance	Size

(c) Label axes and legends [3]

X-axis:

Study Hours

Y-axis:

Score

Colour legend:

Dept

CSE

ECE

MECH

Size legend:

Attendance

70

80

90

A suitable R implementation would be:

```
library(ggplot2)

df <- data.frame(
  ID = c("S1","S2","S3","S4","S5","S6","S7","S8"),
  StudyHours = c(10,5,2,2,4,2,9,8),
  Score = c(95,80,60,85,75,80,85,80),
  Dept = c("CSE","ECE","MECH","MECH",
           "CSE","CSE","MECH","MECH"),
  Attendance = c(80,90,80,70,90,80,90,70)
)

ggplot(df, aes(x = StudyHours,
               y = Score,
               colour = Dept,
               size = Attendance)) +
  geom_point() +
  labs(
    x = "Study Hours",
    y = "Score",
    colour = "Department",
    size = "Attendance"
  )
```

(d) Why mapping Attendance to shape is poor [2]

Attendance is a **continuous variable**.

Shape is better suited to **categorical variables** with only a few categories. Using different shapes for many attendance values would be difficult to distinguish and would make the graph confusing.

Therefore:

Attendance → **size** is better than **Attendance** → **shape**.

Implementation:

